

Motion in a Straight Line

Motion: - Motion is the change in position of an object with time taking a reference point.

→ Motion in a straight line (1-dimensional) is also called as rectilinear motion.

Distance

* The Actual length of path covered by a body is called distance.

* It is a scalar quantity.

* It can never be zero, or negative.

Displacement

* Displacement is the shortest distance in initial and Final position.

+ Displacement = final position - initial position

$$\Delta x = x_2 - x_1$$

+ It is a vector quantity.

+ It can be zero when initial and final position is same.

Average speed

+ Average speed = $\frac{\text{Total path length}}{\text{Total time interval}}$

+ SI unit = m/s or ms^{-1}

+ Always positive.

+ scalar quantity.

Average velocity

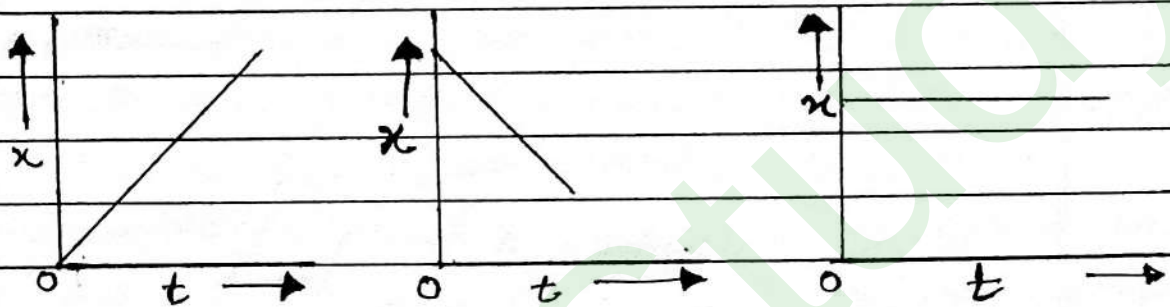
* It is defined as the change in position or displacement (Δx) divided by time interval (Δt), in which displacement occurs.

$$\bar{v} = \frac{x_2 - x_1}{t_2 - t_1} = \Delta x$$

- * Velocity can be positive or negative.
- * It is a vector quantity.
- * SI units = m/s or ms^{-1} .

Avg. speed $\neq$ Avg. velocity

Graphical Representation of v in $x-t$ graph



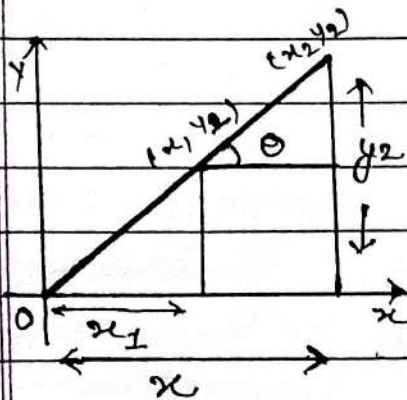
+ve velocity

-ve velocity

Body at Rest

- * Slope of position - time graph gives velocity.

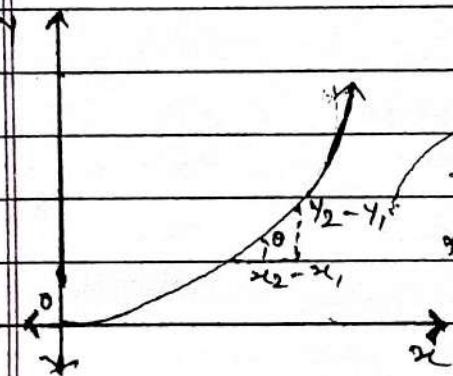
Differentiation



$$\text{Slope} = \tan \theta = \frac{P}{B} = \frac{y_2 - y_1}{x_2 - x_1}$$

→ $\tan \theta = \text{constant}$

* same at every point.

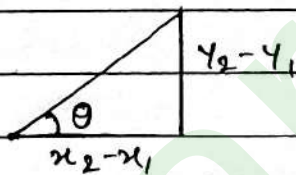


$\tan \theta \Rightarrow \text{variable}$

* vary at every point.

$y_2 - y_1 = 0$ (approx)

$x_2 - x_1 = 0$ (approx)



(The value of a quantity which is very, very small is denoted by dx in physics)

$$\tan \theta = \frac{y_2 - y_1}{x_2 - x_1} = \frac{dy}{dx} \quad (\text{Rate of change of } y \text{ w.r.t } x)$$

y

Formula

1. If y is a constant no. then $\frac{dy}{dx} = 0$

2. If $y = x^n$, then $\frac{dy}{dx} = nx^{n-1}$ e.g. $y = x^3$
 $\frac{dy}{dx} = 3x^2$

3. If $y = x$, then $\frac{dy}{dx} = 1$

e.g. $y = 5x^3$
 $\frac{dy}{dx} = 5 \times 3x^2 = 15x^2$

4. If $y = e^x$, then $\frac{dy}{dx} = e^x$

5. If $y = \log_e x$, then $\frac{dy}{dx} = \frac{1}{x}$

6. If $y = a^x$, then $\frac{dy}{dx} = a^x \log_e a$

7. If $y = \sin x$, then $\frac{dy}{dx} = \cos x$

8. If $y = \cos x$, then $\frac{dy}{dx} = -\sin x$

9. If $y = \tan x$, then $\frac{dy}{dx} = \sec^2 x$

10. If $y = \sec x$, then $\frac{dy}{dx} = \sec x \tan x$

11. If $y = \operatorname{cosec} x$, then $\frac{dy}{dx} = -\operatorname{cosec} x \cdot \cot x$

12. If $y = \cot x$, then $\frac{dy}{dx} = -\operatorname{cosec}^2 x$

Instantaneous velocity and Speed.

* It is the rate of change of position with respect to time at that instant.

* In this, time is very small. (Δt)

$$* \quad V = \frac{\Delta S}{\Delta t} = \frac{ds}{dt} \Rightarrow \boxed{V_{inst} = \frac{ds}{dt}}$$

$$* \quad \boxed{V_{avg} = \frac{S_2 - S_1}{t_2 - t_1}}$$

Que $S = t^3 - 6t^2$, Find velocity at $t = 1s$, $t = 3sec$.
Soln $V = \frac{ds}{dt} = 3t^2 - 12t$ (differentiation)

$$\text{At } t = 1s, v = 3(1)^2 - 12(1) = -9m/s$$

$$\text{At } t = 3s, v = 3(3)^2 - 12(3) = -9m/s$$

Que $S = t^3 - 6t^2$, Find the time when body is doing rest.

$$V = 0$$

Soln $V = \frac{ds}{dt} = 3t^2 - 12t$

$$0 = 3t^2 - 12t$$

$$0 = 3t(t - 4)$$

$$\boxed{t = 0} \quad \boxed{t = 4s}$$

Ques $S = 3t^2 + 4t + 9$. find velocity at $t = 2s$.
And avg. vel. between $1s$ to $4s$.

$$V = \frac{ds}{dt} = 6t + 4$$

$$t = 2$$

$$V = 12 + 4$$

$$V = 16 \text{ m/s}$$

$$\text{Avg. vel.} = \frac{s_4 - s_1}{t_2 - t_1}$$

$$= \frac{73 - 16}{4 - 1}$$

$$= \frac{57}{3}$$

$$= 19 \text{ m/s}$$

$$t = 4.$$

$$s_4 = 3(4)^2 + 4(4) + 9$$
$$= 73$$

$$t = 1$$

$$s = 3 + 4 + 9$$

$$= 16$$

Acceleration

(Rate of change of velocity with time)

- * Average acceleration ($\bar{a}$) over a time interval is defined as the change of velocity divided by time.

$$\bar{a} = \frac{V_2 - V_1}{t_2 - t_1} = \frac{\Delta V}{\Delta t}$$

- * SI unit of acceleration is m/s^2

* $a_{\text{inst}} = \frac{dv}{dt}$

Slope of v-t graph give acceleration

Que $V = 2t^2 - 4t$. Find ① Acceleration at $t = 3\text{sec}$,
② avg. acceleration from 0 to 3s.

Sol

① $a = \frac{dv}{dt} = 4t - 4$

$t = 3$

$a = 4 \times 3 - 4$

$= 12 - 4$

$a = 8 \text{ m/s}^2$

② $t = 0 \quad V = 0$

$t = 3\text{sec}, V_3 = 2 \times (3)^2 - 4(3)$

$= 18 - 12 = 6$

Avg. acceleration = $\frac{V_3 - V_0}{t_3 - t_1}$

$= \frac{6 - 0}{3 - 0} = 2 \text{ m/s}^2$

Que $S = t^3 - 6t^2$. Find the acceleration when body is at rest.
($v=0$)

Solve $V = \frac{dS}{dt} = 3t^2 - 12t$ If $t = 0$
 $a_{\text{ms}} = 6(0) - 12 = -12 \text{ m/s}^2$

$$a = \frac{dv}{dt} = 6t - 12$$

If $t = 4$

$$0 = 3t^2 - 12t$$

$$a_{\text{ms}} = 6(4) - 12$$

$$0 = 3t(t-4)$$

$$= 24 - 12$$

$$\boxed{t=0} \quad \boxed{t=4}$$

$$= 12 \text{ m/s}^2$$

Integration (Reverse process of differentiation)

* Differentiation gives slope of $y-x$ graph.

* Integration gives area of $y-x$ graph
differentiated



* Nearly all functions can be
but some functions can be integrated.

Basic Formula

$\int_a^x \rightarrow$ definite integration

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$ $\int \rightarrow$ Indefinite

2. $\int x^{-1} dx = \log_e |x| + C$

3. $\int e^x dx = e^x + C$

$$S = \int v dt$$

$$v = \int a dt$$

4. $\int dx = x + C$

5. $\int \cos x dx = \sin x + C$

6. $\int \sin x dx = -\cos x + C$

7. $\int \sec^2 x dx = \tan x + C$

8. $\int \csc^2 x dx = -\cot x + C$

$$9. \int \sec x \tan x \, dx = \sec x + C$$

$$10. \int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C$$

$$11. \int \tan x \, dx = \sec x + C \text{ or } -\cos x + C$$

$$12. \int \cot x \, dx = \sin x + C$$

$$13. \int \sec x \, dx = (\sec x + \tan x) + C$$

$$14. \int \operatorname{cosec} x \, dx = -(\operatorname{cosec} x + \cot x) + C.$$

Algebra of Integration

Addition / Subtraction

(Integrate separately)

e.g. $y = x^2 + \frac{1}{x}$

$$\begin{aligned}\int y dx &= \int \left(x^2 + \frac{1}{x} \right) dx \\ &= \int x^2 dx + \int x^{-1} dx \\ &= \frac{x^3}{3} + \log |x| + c\end{aligned}$$

Multiplication

↓

$$\int uv = u \int v dx - \int \left[\frac{du}{dx} \right] \int (v dx) dx$$

e.g. $\int (x \sin x) dx$

I LATE principle tells which one is u

$$\int uv = x \int \sin x dx - \int \left(\frac{dx}{dx} \right) \int (x dx)$$

$$= x (-\cos x) - 1 (-\cos x) dx$$

$$= -x \cos x + \sin x + c$$

e.g. $\int \log x$ or $\int \overset{v}{1} \overset{u}{\log x}$

$$\int 1 \log x = \log x \int 1 dx - \int \left(\frac{d(\log x)}{dx} \int 1 dx \right) dx$$

$$= \log x \cdot (x) - \left(\frac{1}{x} \times x \right) dx$$

$$\boxed{\int \log x = x \log x - x}$$

$$3. \quad v^2 = u^2 + 2as$$

$$a = \frac{dv}{dt} \times \frac{ds}{ds}$$

$$= \frac{dv}{ds} \times \frac{ds}{dt}$$

$$\boxed{a = \frac{dv}{ds} \times v} \quad \text{imp.}$$

$$\int_0^s a \, ds = \int_u^v v \, dv$$

$$a \int_0^s ds = \left(\frac{v^2}{2} \right)_u^v$$

$$a (s - 0) = \frac{v^2}{2} - \frac{u^2}{2}$$

$$a (s - 0) = \frac{v^2 - u^2}{2}$$

$$2as = v^2 - u^2$$

$$\Rightarrow v^2 = u^2 + 2as$$

★ Always correct

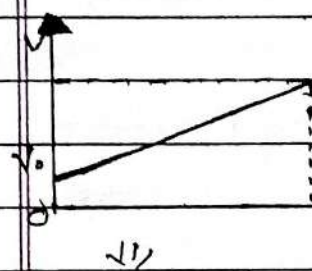
$$a = \frac{dv}{dt}$$

$$v = \frac{ds}{dt}$$

$$a = \frac{v \, dv}{ds}$$

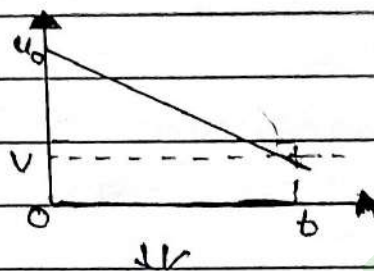
Displacement-time graph / velocity-time graph / acceleration-time graph

1. Velocity-time Graph

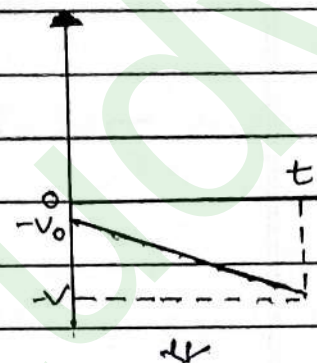


motion in a +
direction

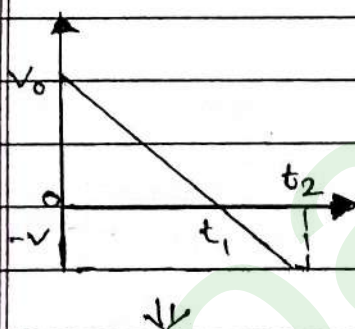
+ve acceleration
(constant)



+ve direction,
-ve acceleration
(uniform
retardation)



-ve direction,
-ve acceleration



-ve acceleration
changing direction.

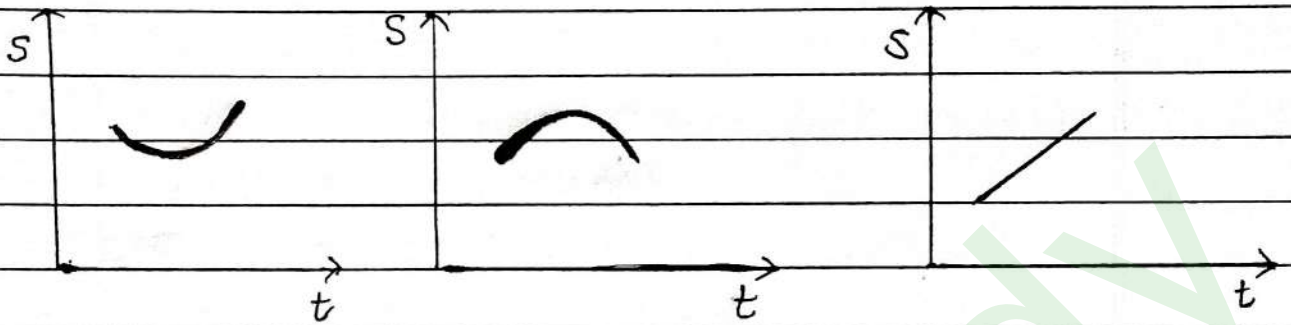
$\theta < 90^\circ \rightarrow$ Slope +ve.

$\theta > 90^\circ \rightarrow$ Slope -ve.

* Slope of v-t graph - acceleration

* Area under v-t curve gives displacement.

2. Displacement (Position) - time graph

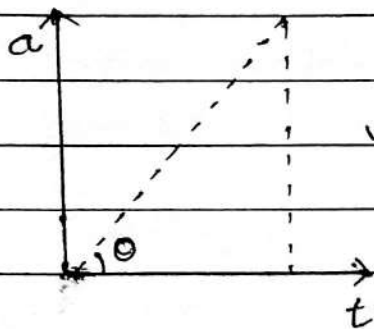


concave upward concave downward Zero acceleration
 $a \rightarrow +ve$ $a \rightarrow -ve$

If, acceleration } same sign $\rightarrow$ speed increases
velocity $\downarrow$ $(++; --)$

If different sign $\rightarrow$ speed decreases.

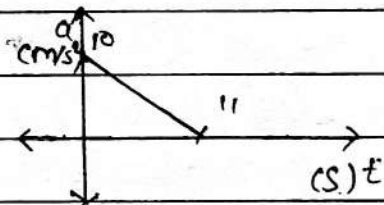
3. Acceleration - time Graph



$$\text{Slope} = \frac{\Delta a}{\Delta t} = \text{ jerk } (m/s^3)$$

+ Area under a-t graph gives change in velocity.

Ques



Find change in velocity
From the given graph

$$\begin{aligned} \text{Velocity} &= \text{Area of a-t graph} \\ &= \frac{1}{2} \times 11 \times 10^5 \end{aligned}$$

$$\text{change in velocity} = 55 \text{ m/s}$$

Uniformly increasing velocity $\rightarrow$ acceleration constant.

non-uniformly increasing velocity $\rightarrow$

acceleration increase uniformly

Relative Velocity (1-D)

- + Relative velocity depends on the frame of reference
↓

Place where observer stands

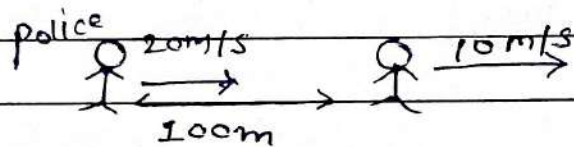
- + Relative velocity / acceleration / displacement depends upon the frame of references

- + Relative distance and time does not depend on frame of reference.

- + velocity of object A relative to object B is

$$V_{AB} = V_A - V_B$$

Que A policeman runs to catch a thief.
The velocity of police and thief are 20 m/s and 10 m/s respectively.
Find the relative velocity of police to thief if the distance b/w them is 100 m and also time.



$$V_{pt} = V_p - V_T$$
$$= (20 - 10) \text{ m/s}$$

$$V_{pt} = \underline{10 \text{ m/s}}$$

$$S (\text{distance}) = 100 \text{ m}$$

$$\text{Speed} = \frac{d}{t}$$

$$t = \frac{d}{v} = \frac{100}{10} = \underline{10 \text{ s}}$$

Que If two stones are dropped from a tower with the velocity of 0 m/s and 3 m/s. Find the separation between the stones after 3s.

Soln In motion under gravity (1-D)
 $\Rightarrow a_{AB} = 0$ (always)

$$V_{AB} = V_A - V_B$$

$$= 0 - (-3) \text{ (-ve sign is due to force in downward direction)}$$

$$= +3 \text{ m/s}$$

$$a_{AB} = 0$$

$$t = 3 \text{ second}$$

$$\text{separation (distance)} = v \times t$$

$$= 3 \times 3$$

$$= \underline{\underline{9\text{m}}}$$

$\Rightarrow$ For a body thrown vertically up with velocity u .

- maximum height reached (h) = $\frac{u^2}{2g}$
- Total time of flight = $\frac{2u}{g}$
- Velocity attained by a body dropped from height h ; $v = \sqrt{2gh}$.
- Time of ascent = Time of descent = $\frac{u}{g}$